LUCILA VÁZQUEZ SORIANO

Junior Data Scientist | Machine Learning · NLP · Deep Learning

Alicante, Spain | lucilavsoriano@gmail.com | linkedin.com/in/lucilavazquezsoriano | lvs42.github.io

SUMMARY

Junior Data Scientist with an MSc in Data Science (2026) and an engineering degree. Machine learning and deep learning projects in NLP, computer vision and causal inference. Seeking roles in NLP and Large Language Models (LLMs).

EDUCATION

MSc in Data Science, University of Alicante | Sep 2025 – Sep 2026

- Thesis: deep learning for gastric cancer histopathology. Grade: 10/10.

BSc in Multimedia Engineering, University of Alicante | Sep 2019 – Jan 2025

- Thesis: VR escape rooms to prevent cognitive decline. Grade: 10/10. Published at SOCO 2025 (Springer).

PROJECTS

NLP: Online Sexism Detection (SemEval 2023 Task 10) | Python, Hugging Face Transformers, scikit-learn, Gradio

- Achieved 0.80 macro-F1 on binary sexism detection with DistilBERT.
- Beat the best TF-IDF baseline (SVC, 0.73) by 7.5 pp after testing class-balancing strategies.
- Built a Gradio demo for interactive testing.

Deep Learning: Gastric Cancer Classification (MSc Thesis) | PyTorch, CatBoost, SHAP, Docker

- Reached 94.70% accuracy (+1.5 pp) by adding CBAM attention to EfficientNetV2-B0.
- Validated on 245,196 images with stratified 5-fold cross-validation and 5 random seeds.
- Explained predictions with SHAP and attention maps.

Production ML: Serverless People-Detection Pipeline | Azure Functions, Blob Storage, Azure AI Vision

- Deployed an event-driven pipeline that processes one frame every 0.5 seconds.
- Automated publication of detection results to a static web app.

Statistics: Causal Inference on Urban Cycling (London) | Causal Forest, Propensity Score Matching

- Found that bike lanes raise cycling and steep slopes reduce it, using Causal Forest and PSM.
- Built a 2008–2024 dataset from official counts, Street View imagery and OpenStreetMap.

Additional projects | TensorFlow, Keras, PyTorch Geometric, GeoPandas

- **Model calibration:** reduced ECE by 86% on Fashion-MNIST with Focal Loss and Temperature Scaling.
- **Graph neural networks:** raised MUTAG accuracy from 74.0% to 79.4% with Laplacian positional encodings.
- **Geospatial analytics:** built a “15-minute city” accessibility index for Madrid in a 4-person team.

EXPERIENCE

Software Developer Intern, Doña Rosa S.L. | Oct 2023 – Dec 2023

- Built real-time client-server communication in C++ with Socket.IO and Unreal Engine.
- Migrated projects across engine versions and tested web applications.

Student Guide and Representative, University of Alicante | Dec 2022 – Apr 2023

SKILLS

- **Programming:** Python, SQL, R, C++, Java, JavaScript
- **Machine Learning:** scikit-learn, CatBoost, classification, cross-validation, model calibration
- **Deep Learning & NLP:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Natural Language Processing (NLP), CNNs
- **Statistics:** probability, Bayesian methods, causal inference, count models
- **MLOps & Cloud:** Docker, Git, Azure, AWS, APIs, ETL

CERTIFICATIONS

- Generative AI Foundations, AWS (IMMUNE): prompt engineering, responsible AI

PUBLICATIONS

- VR Escape Room to Prevent Cognitive Decline in Older Adults. SOCO 2025, Springer. Co-author.

LANGUAGES

- Spanish (native), English (C1), Catalan (C1), Japanese (A2)